

Quick Sort Algorithm

What is it ?

- Divide and Conquer: split input into smaller chunks, solving those subsets, repeated until reaching a fundamental unit that can be easily solved
- 'pivot': an elements used to partition the array with less than or equal on one side, greater than on the other side
- $n \log n$ (best case)($\log n$ amount of n) to n^2 (worst case)
- worse case when sorting a reversed sorted array, choosing pivot make unbalanced partition

Why it work ?

- QuickSort recursively divides the array by selecting a pivot, partitioning elements around the pivot, and then applying the same process to the subarrays until the entire array is sorted.